HENRY KOU

kenryhou2.github.io | hkou@andrew.cmu.edu | (831) 233-4058 | US Citizen | linkedin.com/in/henrykou

Education

Carnegie Mellon University (CMU) — *Pittsburgh, PA, USA*
Master of Science in Robotics, School of Computer Science — 2024 - 2026
Thesis Committee: Dr. Howie Choset, Dr. Wennie Tabib, Dr. Darwin Mick

University of California, Los Angeles (UCLA) — *Los Angeles, CA, USA*
Bachelor of Science in Electrical Engineering — 2017 - 2021
Advisor: Dr. William Kaiser

Selected Publications

- Shricharan Shastry, Darwin Mick, **Henry Kou**, Yuchen Wu, Shankruth Balasubramaniyam, Junil Min, Lu Li, Anthony Baker, and Howie Choset. "*Reactive Deposition-Event Scheduling for Translational Vibration Compensation in Discrete Material Deposition Processes*." IEEE Robotics and Automation Letters, RA-L, 2027 (preprint).
- Zhikai Zhang, Siqi Guo, **Henry Kou**, Ishayu Shikhare, Howie Choset, and Lu Li. "*Bio-Inspired Distributed Neural Locomotion Controller (D-NLC) for Robust Locomotion and Emergent Behaviors*." IEEE International Conference on Robotics and Automation (ICRA), 2025. [Link]

Research & Work Experience

Carnegie Mellon University, Robotics Institute — *Pittsburgh, PA, USA*
Biorobotics Lab, Staff Researcher — 2022 – Present.
Research advisors: Dr. Howie Choset, Lu Li

- Designed and implemented a real-time robotic perception and process-control system in ROS2 for aerospace inkjetting using a UR5e manipulator, integrating multi-rate IMU and motion-capture sensor fusion with GTSAM factor-graph optimization, fixed-lag smoothing, and inertial preintegration for 1 kHz 6-DoF state estimation.
- Developed a novel reactive and predictive discrete deposition-event scheduling algorithm using real-time tool-state estimation and motion prediction to compensate for vibration-induced spatial errors, achieving over 85% reduction in mean deposition error compared to fixed-time scheduling.
- Developed embedded IMU sensor processing on PSoC microcontrollers for the 6-Legged EigenBot, a modular robotic platform, enabling proprioceptive neural feedback for stable locomotion. Trained reinforcement learning-based locomotion policies in NVIDIA IsaacLab to create stable, robust gaits.

Texas Instruments — *Santa Clara, CA, USA*
Analog Circuit Applications Engineer, High-Current Power Regulators — 2021 – 2024

- Designed and debugged multiphase buck-converter hardware and PCB design, including efficiency, EMI, and closed-loop performance; contributed to $15M+ in custom power-design projects for Tesla, Xilinx, and Alpine with technical publication.

Raytheon Technologies — *Los Angeles, CA, USA*
Digital Hardware Intern — 2020

- Supported digital hardware design and testing, including power and thermal evaluation of aerospace programmable logic and FPGA-based systems.

Naval Postgraduate School, Center for Autonomous Vehicles Research — *Monterey, CA, USA*
Robotics Research Intern — 2018, 2019

- Simulated autonomous underwater vehicle navigation in ROS1 using SLAM and GPS-denied inertial odometry.

Relevant Skills

Theoretical Foundations: Robot SLAM, Sensor Fusion (Batch and Recursive methods), IMU-Preintegration, Optimal Control and Reinforcement Learning, Multi-Agent Path Planning, Linear Feedback Systems, Digital Signal Processing, Linear Algebra, Differential Equations. Independent Study: Lie Groups and Manifolds

Programming & robotics: C, C++, Python, ROS, Linux, GTSAM, Git, MATLAB, Julia, MuJoCo, Isaac Lab

Hardware: Teensy, STM32, MSP430, PSoC, Raspberry Pi, Altera FPGAs, VHDL, SystemVerilog, SPI, I2C, Altium Designer, switching power regulators, SIMPLIS, PSpice, Mbed

Service

Volunteer — IEEE/RSJ International Conference on Intelligent Robots and Systems. Pittsburgh, PA (2026)

Research Mentorship — Mentored undergraduate and graduate students, CMU Biorobotics Lab (2024–2026)

Projects

EigenBot — Modular Robot for Rapid Prototyping
Research collaboration
- Contributed to a modular robotic limb platform integrating force sensing, embedded hardware, and distributed neural locomotion control using proprioceptive feedback and central pattern generators. eigenbot-dnlc.github.io.

Twin Turbo Multiphase Board — Reference Design
Power electronics hardware project
- Designed, assembled, and tested a stackable TI buck-power reference design supporting up to six phases; benchmarked conversion efficiency, load transient response, and EMI.

Micromouse — Autonomous Maze-Solving Robot
Embedded robotics competition project 2019
- Competed in collegiate Micromouse events; earned first place in the first-year category at UCLA and third place at UCSD in 2019.

Awards & Honors

- UCLA IDEAHacks — 2nd Place, Gnome (2021)
- UCLA IDEAHacks — Honorable Mention, Salt-o-matic (2020)
- California Micromouse Competition, UC San Diego — 3rd Place, Team HITB (2019)
- All-American Micromouse Competition, UCLA — 1st Place, First-Year Category, Team O (2019)
- Silicon Valley Regional, FIRST Robotics Competition Team 2035 — Chairman's Award (2016)
- Silicon Valley Regional, FIRST Robotics Competition Team 2035 — Engineering Inspiration Award (2014)

Technical Skills

Programming & robotics: C, C++, Python, ROS, Linux, GTSAM, Git, MATLAB, Julia, MuJoCo, Isaac Lab

Embedded & digital systems: Teensy, STM32, MSP430, PSoC, Raspberry Pi, Altera FPGAs, VHDL, SystemVerilog, SPI, I2C

Electronics & circuit simulation: Altium Designer, switching power regulators, SIMPLIS, PSpice, Mbed

Relevant Coursework

Robot Localization and Mapping; Optimal Control and Reinforcement Learning; Multi-Agent Path Planning; Linear Feedback Systems; Digital Signal Processing; Computer Architecture and Operating Systems; Advanced Circuit Analysis.

Other Publications

- Yi Yang, Xiang Fei, Lehong Wang, Zilin Dai, Ruogu Li, Jiting Cai, Liyao Chang, Xinyi Yang, **Henry Kou**, Ruijie Fu, Lu Li, and Howie Choset. "*Point It, Strike It: Direction-Conditioned Dynamic Manipulation of Deformable Linear Objects*." ICRA 2027 submission (preprint), 2026.
- Y. Yang, X. Fei, L. Wang, C. Li, Z. Dai, H. Kou, L. Li, and H. Choset, "DeformX: A Versatile Co-Simulation Framework for Deformable Linear Objects," in Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Link]
- Geordan Gutow, Ben Spin, Anoop Bhat, Andrew Orekhov, Beau Pollard, Ye Jin, Ralph Boirum, **Henry Kou**, Shricharan Shastry, Burhanuddin Shirose, Matt Travers, and Howie Choset. "*Hardware-in-the-Loop Evaluation of Admittance Controller for Satellite Payload Insertion by a Free-Floating Manipulator*." IEEE journal submission (preprint), Dec. 2024.

Texas Instruments Technical Publications
- **Henry Kou** and Patrick Shalton. "*EVM User's Guide: TPSM64406EVM 36-V, Dual 3-A Output Synchronous Buck Module*." Texas Instruments, Nov. 2023. [Link]
- Eric Lee and **Henry Kou**. "*User's Guide: LMQ644A2-Q1 6-Phase Buck Regulator Design for Automotive ADAS Applications*." Texas Instruments, July 2023. [Link]
- Marc Davis-Marsh and **Henry Kou**. "*TPSM64404/06 3-V to 36-V, Low IQ, Dual 2/3-A Module Optimized for Power Density and Low EMI*." Texas Instruments, Dec. 2023. [Link]
- **Henry Kou** and Jimmy Hua. "*User's Guide: LMR54450-Q1 36-V, 5-A Buck Converter Evaluation Module*." Texas Instruments, June 2022. [Link]